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Field Safety, Wellbeing, and Productivity: A Practical Model for Empowering Oilfield Teams

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Abstract

This paper presents a replicable, field-based approach to enhancing safety culture, employee wellbeing, and team productivity in a mature oilfield located in the Amazon region. The objective is to demonstrate how grassroots initiatives targeting physical health, psychological wellness, and proactive HSE behavior can empower operational teams while strengthening overall safety performance.

In early 2024, a local group of employees, created for supporting organizational HSE activities, launched two integrated programs in a remote field setting. The first initiative, "Live Well," focused on improving physical and emotional wellbeing through inclusive activities such as sports tournaments, functional fitness sessions, Olympic-style team events, and collaborative games. These efforts were aimed at fostering engagement, mental resilience, and physical health across all levels of personnel. They also promoted cross-departmental integration and healthy competitiveness by uniting all teams around a common goal: succeeding in the organized HSE challenges. The second initiative, "Hazards Hunt," promoted safety ownership via cross-functional inspections, peer-to-peer accountability, and environmental stewardship. Field teams and supervisors jointly identified risks, recognized safe behaviors, and implemented corrective actions. A structured recognition system rewarded safety leadership and environmental best practices.

Results from the combined programs included over 1,700 man-hours of wellness activities, the identification and mitigation of 153 safety risks, control of 104 environmental aspects, and the formal recognition of more than 62 safe behaviors and 23 environmental initiatives. The initiatives also led to improved morale, increased participation from supervisors and contractors, and stronger collaboration across departments. Teams previously disengaged became active contributors to well-being and risk prevention efforts, resulting in reduced reactive interventions and a more preventive safety culture.

This paper introduces grassroots, people-centered model for cultivating safety and wellbeing in remote, high-risk environments. By blending wellness, leadership development, and operational HSE integration, the model offers a scalable framework that enhances field-level resilience, inclusiveness, and motivation—key elements for building the energy workforce of the future.

Introduction

In recent years, the energy industry has acknowledged the growing importance of integrating employee wellbeing, psychological safety, and leadership development into core operational strategies. However, translating this awareness into practical, field-level actions remains a significant challenge, especially in remote and high-risk environments.

This project was developed for one of the largest oil-producing fields in Ecuador, with 260 active wells including oil and water producers located in the Orient Basin. (Fig. 1), located in the Amazonian region of Ecuador (Auca – Block 61), represents a complex operational landscape where over 130 personnel support approximately 250 wells, delivering 72,000 barrels of oil per day. The field operates under demanding conditions: geographic isolation, high environmental sensitivity, multicultural teams, and the need for consistent, incident-free performance. In such settings, conventional top-down approaches to safety and engagement often fall short, struggling to resonate with frontline workers or adapt to the nuances of daily operations.

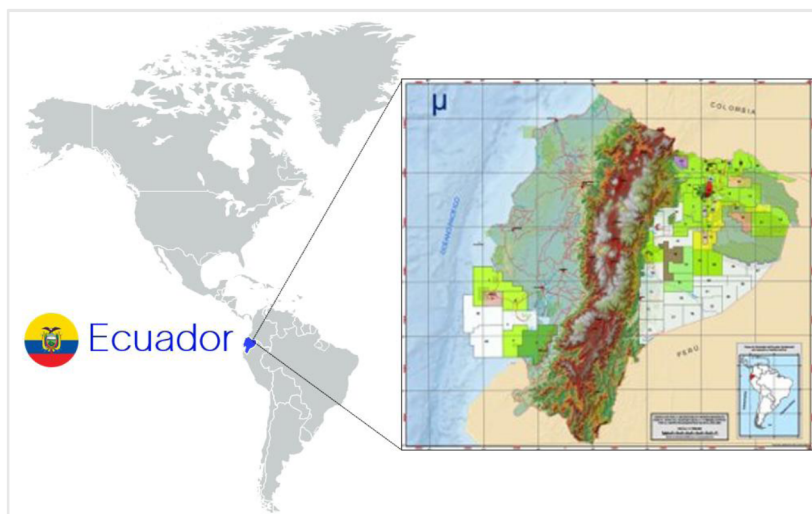


Figure 1—Remote field location map of the Amazon region.

Also, in the context of a high-risk industry, specifically hydrocarbon production—it is essential to ensure that processed barrels meet safety standards. This implies that the entire production process involving the client, the organization, and external contractors must be conducted within a safe work environment that also promotes environmental stewardship and the overall well-being of personnel.

The organizational goal is to prevent accidents and maintain personnel within a safe environment. For this reason, an HSE (Health, Safety, and Environment) team has been established to carry out proactive activities aimed at involving all personnel to prevent personal incidents. However, daily operations often limit the active participation of operational staff in safety and environmental programs, leaving their promotion solely to the HSE function. To achieve better integration between operations and HSE support, the organization has created a group of employees tasked with empowering operational teams on key HSE topics, serving as additional support in the pursuit of zero accidents within a pleasant and safe working environment.

In response to this reality, a group of field personnel supervisors with the support of HSE staff created a solution from within: the Organizational Empowerment Team. Rather than being a corporate initiative imposed from headquarters, it emerged as a grassroots platform to empower teams, connect leadership with the field, and embed a culture of care and accountability.

Structured around three key pillars—Talk, See, and Check—ET functions as a living system for continuous HSE engagement, situational insight, and compliance ownership. It has become a focal point

for workforce-driven innovation, designed to align safety culture with the human experience of working in the field.

This paper outlines the journey through various activities, offering insights into how field-led initiatives by operational personnel can transform high-risk and remote oilfield operations into safe working environments that prioritize employee well-being and environmental responsibility. Through proactive risk management across different job sites—and above all, by integrating these activities into the production process—this approach demonstrates how worker well-being can be harmonized with operational goals, encouraging inclusive participation across the entire workforce. By documenting the rationale, structure, and outcomes of the team's actions, this work seeks to provide a scalable framework for organizations aiming to enhance both performance and well-being in complex field environments.

Statement of Theory and Definitions

The theoretical foundation of this paper is rooted in the principles of participatory safety culture, holistic employee wellbeing, and behavioral safety leadership. These principles guide the structure and implementation of the Organizational Empowerment Team (OET) model.

At its core, OET is designed as an action-oriented mechanism that empowers field personnel to take ownership of their health, safety, and environmental performance. Rather than functioning as a compliance-driven system, OET creates a collaborative platform where employees, line management, and the HSE function work together to improve operational culture and performance. This approach reflects a bottom-up philosophy, where those closest to operational risks are enabled to lead positive change.

The OET model is organized into three functional pillars:

- **Talk (HSE Engagement):** Focuses on driving emotional connection, shared purpose, and workforce communication through structured campaigns, safety meetings, and interactive activities.
- **See (HSE Situational Insight):** Enhances hazard recognition and proactive intervention by supporting training compliance, risk assessments, and on-site observations. The See group serves as the analytical backbone of the initiative.
- **Check (HSE Compliance):** Reinforces systematic performance through audits, inspections, and remedial action planning. This group ensures that HSE strategies are sustainable and aligned with regulatory and internal expectations.

Two core programs were developed within this framework:

- **"Live Well":** Based on the concept of holistic wellbeing, Live Well incorporates physical fitness, emotional resilience, and team bonding into a structured health promotion model. It reflects emerging evidence that personal wellbeing is directly linked to operational effectiveness, safety compliance, and long-term engagement in high-risk industries.
- **"Hazards Hunt":** This program is rooted in Behavior-Based Safety (BBS) theory, which holds that reinforcing positive behaviors and peer accountability can lead to significant reductions of incidents. Hazards Hunt encourages multidisciplinary teams to identify, report, and correct unsafe conditions, while recognizing safe behaviors and environmental stewardship through a structured reward system.

Key terms used throughout this paper include:

- **Behavior-Based Safety (BBS):** A systematic approach that promotes safe behavior through observation, feedback, and positive reinforcement.
- **Wellbeing (in the field context):** The physical, emotional, and social condition of personnel, recognized as a performance enabler in remote, high-risk operations.

- Proactive Safety Culture: An organizational mindset where individuals anticipate, report, and address risks before they result in incidents.
- Cross-functional Team Participation:

A core strategy for strengthening preventive management. The variety of specialties allows potential hazards to be addressed from multiple angles, enriching the analysis of unsafe conditions and facilitating the design of more effective corrective actions. Furthermore, this integration promotes a safety culture grounded in collective commitment, technical knowledge-sharing, and the pursuit of innovative solutions. Ultimately, it supports both the reduction of workplace incidents and the optimization of operational processes in safe and productive environments.

By integrating these theoretical models with practical, field-driven action, the OET framework enables a scalable and adaptable pathway to safety, resilience, and workforce motivation in challenging environments.

Description and Application of Equipment and Processes

The implementation of the OET programs at Shaya field— "Live Well" and "Hazards Hunt"—required not only human commitment but also logistical coordination, infrastructure adaptation, and communication systems to support sustained field-level engagement.

Live Well Program Setup and Execution

The Live Well program was developed as a strategic initiative to promote the physical, emotional, and mental well-being of field personnel. It consists of a set of scheduled activities designed for individuals of any age or physical condition. Recognizing the demands of operational environments, this program implements structured recreation and wellness activities aimed at fostering resilience, boosting morale, and supporting overall health. By prioritizing holistic care, Live Well not only helps mitigate the stress and fatigue associated with fieldwork but also contributes to a safer, more engaged, and productive workforce. The program was executed with minimal disruption to operations by scheduling events during off-shift hours (typically after 16:00) and weekends.

Facilities and Equipment Used:

- Existing sports fields and open areas were repurposed for activities such as football, volleyball, and relay races.
- A designated wellness zone was established using a decommissioned warehouse area, where functional training sessions and group exercises were conducted.
- Equipment included:
 - Stretching bands, agility cones, and medicine balls
 - Audio systems for guided workouts and music-based sessions (e.g., "bailoterapia")
 - Six bicycles acquired for aerobic exercise on a designated track
 - Portable hydration and first aid stations for each event

Coordination and Delivery:

- Events were facilitated by trained field personnel and local safety coaches, supported by the Medical and HSE departments.
- Programs were planned bi-weekly, using an internal calendar system shared via Yammer and onsite communication boards.

- Participation was voluntary but promoted through team challenges, inclusive messaging, and recognition awards.

Hazards Hunt Process and Field Application

The Hazards Hunt program was designed to strengthen the HSE culture through structured inspections focused on the proactive identification of risks. This initiative promotes shared responsibility by actively involving operational personnel who do not hold HSE roles within the organization, supported by the HSE function as a key part of the process, and through the collaboration of various field teams who enabled "Hazards hunt" within their operations. This fosters a collaborative environment where each team member actively contributes to continuous improvement. By integrating field observations with dialogue among participants, Hazards Hunt enhances preventive awareness and reinforces the commitment to safe practices across all operations, helping the organization comply with its Internal Safety Regulations and with Ecuadorian local regulations regarding the maintenance of a safe work environment and the corresponding follow-up.

Process Structure:

- Cross-functional teams, including technicians, supervisors, and contractors, were formed for each site walkaround.
- Teams used predefined HSE checklists, based on system, customized to the operational realities of a mature field.

Field Execution:

- Walkarounds were carried out at least every two weeks, covering various operational zones (production pads, maintenance areas, utility zones).
- All findings were logged in the platform, categorized as:
 - Unsafe conditions
 - Unsafe behaviors
 - Environmental aspects

Recognition and Reporting:

- Positive observations (safe behaviors and environmental best practices) were submitted for quarterly recognition under the SAFE program.
- Outstanding contributions were acknowledged during all-hands meetings and shared via field dashboards.

Innovative elements:

- A gamified risk leaderboard was introduced to promote healthy competition between departments, showing which teams identified the most actionable risks.
- Findings were also used to update local training content and toolbox talks, ensuring feedback loops into operational learning.

Evaluation of Tools and Methods

Despite using low-cost, field-available resources, both programs proved highly effective due to their simplicity, adaptability, and team-driven design. No specialized experimental equipment was required; rather, success relied on reframing existing tools (spaces, materials, communication platforms) for

new purposes. The programs' strength lay in the social architecture—clear objectives, visibility, peer accountability, and real-time feedback.

What worked best was the continuity and consistency of the activities. Scheduling events every two weeks and ensuring they took place as planned allowed participants to gradually develop interest and engagement. Team-based games and collaborative dynamics were particularly effective in strengthening interpersonal connections and fostering a healthy sense of competition. Each activity was designed to reinforce an HSE theme, ensuring that key safety principles remained at the center of the initiative. Moreover, leadership support played a critical role: supervisors actively encouraged participation and motivated their teams to be involved.

The main barrier encountered at the beginning was a resistance to shifting mindsets. Team members were initially focused solely on technical tasks, and it was challenging to encourage participation in activities perceived as "non-operational" or outside routine responsibilities.

Adaptability was achieved by designing activities that emphasized teamwork and inclusion. Games and exercises encouraged collaboration between departments, helping participants to connect on a human level.

In the case of "Hazard Hunts", motivation increased when individuals saw clear objectives and recognition for reporting. A particularly innovative aspect was the involvement of personnel from non-production departments—such as facilities, HSE, and logistics—in operational safety observations. This cross-functional participation not only broadened risk awareness but also empowered individuals to contribute their unique perspectives by identifying positive practices and potential hazards outside their usual scope.

These methodologies may serve as scalable templates for other remote operations seeking to embed culture, wellbeing, and proactive safety without disrupting core production goals.

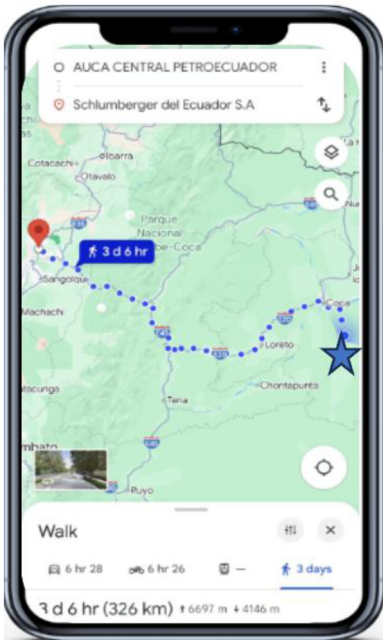
Presentation of Data and Results

The implementation of the "Live Well" and "Hazards Hunt" programs produced quantifiable and qualitative results that demonstrate the effectiveness of the OET model in improving safety engagement, employee wellbeing, and operational cohesion.

Quantitative Results

Live Well Program. As part of the Live Well initiative, during 2024 a total of 10 integration activities were conducted, including group contests, team-based challenges, strategy games, and other collaborative events. Additionally, 80 sports activities were organized, such as 3K and 5K running events, football and volleyball tournaments, and functional training sessions. These efforts resulted in a cumulative total of 1,700 man-hours dedicated to recreation and physical exercise.

To illustrate the scale of engagement, these 1,700 hours are equivalent to 11 round-trip walks between the organization's offices in Quito, Ecuador, and the field offices located in Block 61. The program successfully fostered team integration among field personnel, regardless of age or physical condition, reinforcing inclusivity and holistic well-being. (Pic 1).



Pic. 1—Scale of engagement in a Round-trip to field

Hazards Hunt. During 2024, a total of 19 multidisciplinary field inspections were conducted as part of the Hazards Hunt program. Each session recorded an average participation of 7.8 individuals, in addition to the personnel evaluated from 9 different contractors. These inspections led to the identification and correction of 153 substandard conditions and 104 environmental aspects. In total, 257 risks that had the potential to trigger HSE-related events were eliminated. (Fig. 2)

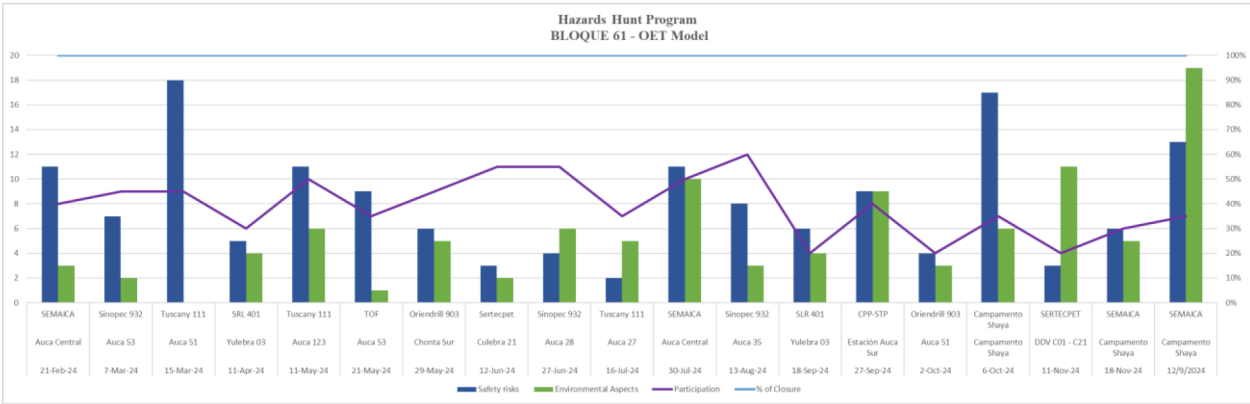


Figure 2—Distribution of substandard conditions and environmental aspects in Hazards Hunt 2024 program and staff participation

To analyze and present the findings, descriptive statistics were applied. This approach allowed for a clear quantification of participation levels, types of findings, and corrective actions taken. The results provide a measurable overview of the program's impact on operational safety and environmental compliance.

In addition to the corrective actions taken, the Hazards Hunt program actively promoted the recognition of individuals who demonstrated a strong commitment to safety culture within their operations. As a result, 62 safe behaviors were identified and formally acknowledged during the various inspections conducted throughout the year. Furthermore, 23 best environmental practices were also recognized during site visits, highlighting proactive efforts to maintain compliance and minimize environmental impact. (Fig. 3).

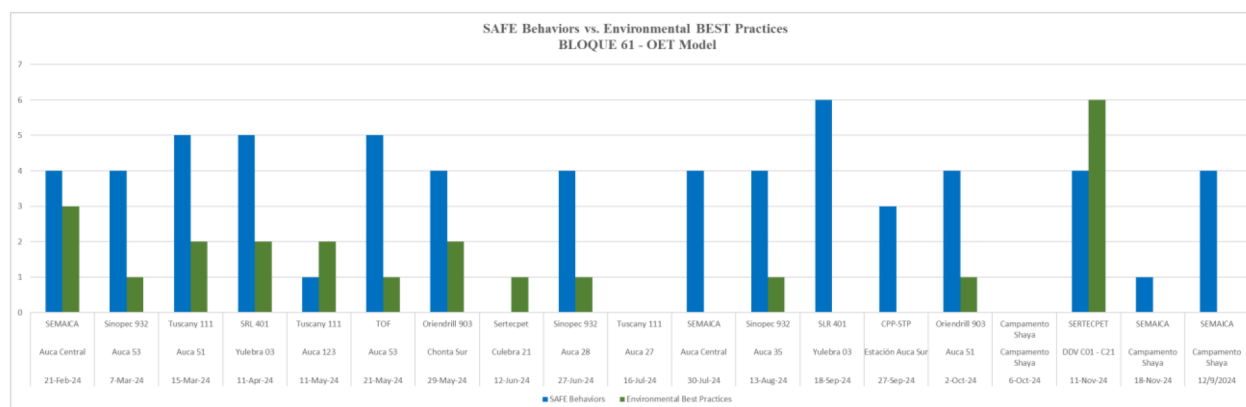


Figure 3—Safety culture and environmental Best practices identified in Hazards Hunt 2024 program.

These recognitions served not only to reinforce positive conduct but also to encourage peer engagement and continuous improvement across operational teams. The identification and documentation of these behaviors and practices were integrated into the program's descriptive statistical analysis, providing qualitative support to the quantitative outcomes.

Across the 19 inspections conducted in 2024, 45% of the findings corresponded to safety risks, while 30% were related to environmental aspects, all of which were corrected during the inspection process. Additionally, the program emphasized the recognition of positive contributions to the HSE culture: 18% of the observations were acknowledged as safe behaviors, and 7% as environmental best practices. (Fig 4).

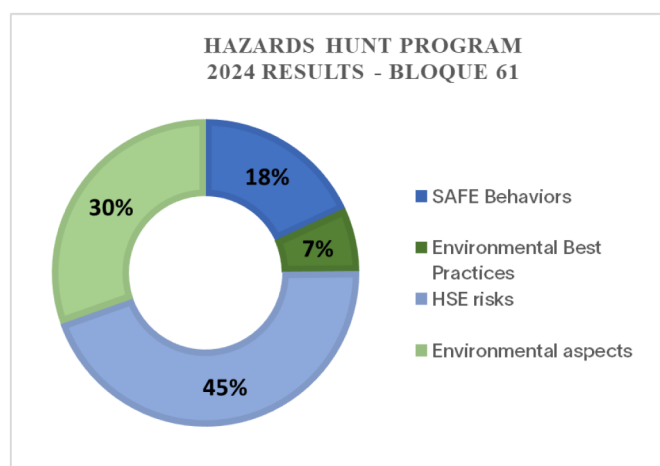


Figure 4—Hazards Hunt 2024 program -findings

This distribution reflects the program's dual focus on risk mitigation and cultural reinforcement, supporting both compliance and continuous improvement. The data were analyzed using descriptive statistics and visualized to highlight the proportional impact of each category.

Observations and Trends

Live Well. Achieved over 70% participation rate among field personnel.

Involved staff previously disengaged from health programs, including supervisors and shift leaders.

Events created positive emotional impact, as noted in field feedback and participation surveys.

Team spirit improved significantly, especially in departments with high operational stress.

Hazards Hunt. Promoted a peer-accountability environment, where personnel took pride in identifying and solving risks.

Enabled joint inspection efforts between organization teams and contractors.

Contributed directly to incident prevention, with zero recordable incidents during inspection weeks.

Outcomes were used to adjust and prioritize HSE toolbox talks, creating a feedback loop from field to planning.

Impact on Field Culture and Engagement

Workers reported higher trust in leadership, due to their visible involvement and recognition practices.

Interdepartmental collaboration improved, as activities required coordination across functional areas.

The atmosphere of inclusiveness and ownership led to the surfacing of previously unreported risks.

Field supervisors became HSE multipliers, replicating practices in daily safety conversations and shift meetings.

These results reflect the potential of grassroots HSE programs to drive measurable cultural and operational improvements, particularly in remote, high-risk environments. The scalability of this model allows for rapid replication in similar field contexts without heavy investment in infrastructure or complex technology.

Conclusions

The deployment of the Organizational Empowerment Team (OET) model at field—through the implementation of "Live Well" and "Hazards Hunt"—demonstrates that field-led, people-centered initiatives can significantly enhance safety performance, employee wellbeing, and operational collaboration.

Key conclusions from this initiative include:

1. **Grassroots engagement drives sustainable cultural change.** When field personnel are given ownership of safety and wellness programs, participation and accountability increase. The OET model enabled individuals at all levels to become active contributors to a proactive HSE culture.
2. **Wellbeing is a performance enabler.** The "Live Well" program showed that structured physical and emotional wellness activities improve morale, increase resilience, and reduce disengagement, particularly in remote and high-pressure environments.
3. **Peer-based safety inspection enhances risk awareness.** The "Hazards Hunt" model encouraged teams to identify, report, and address hazards collaboratively, resulting in early risk mitigation, improved environmental control, and more frequent positive behavior recognition.
4. **Simple tools, when embedded in a strong social structure, outperform complex systems.** No advanced technology or capital investment was required. The success of the programs relied on structured coordination, leadership support, and peer motivation.
5. **The OET framework is scalable and adaptable.** The model can be replicated across other field locations with minimal adjustment, offering a strategic approach to improving safety culture and operational integrity in remote or high-risk environments.

These conclusions reinforce the value of integrating wellbeing and leadership development into operational HSE strategies, not as optional add-ons, but as core drivers of performance and resilience.

Acknowledgments

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Recognition is extended to the operational leadership at **Shaya Field – Block 61**, whose openness to innovation enabled the successful implementation of a grassroots model that is already creating cultural and operational value.